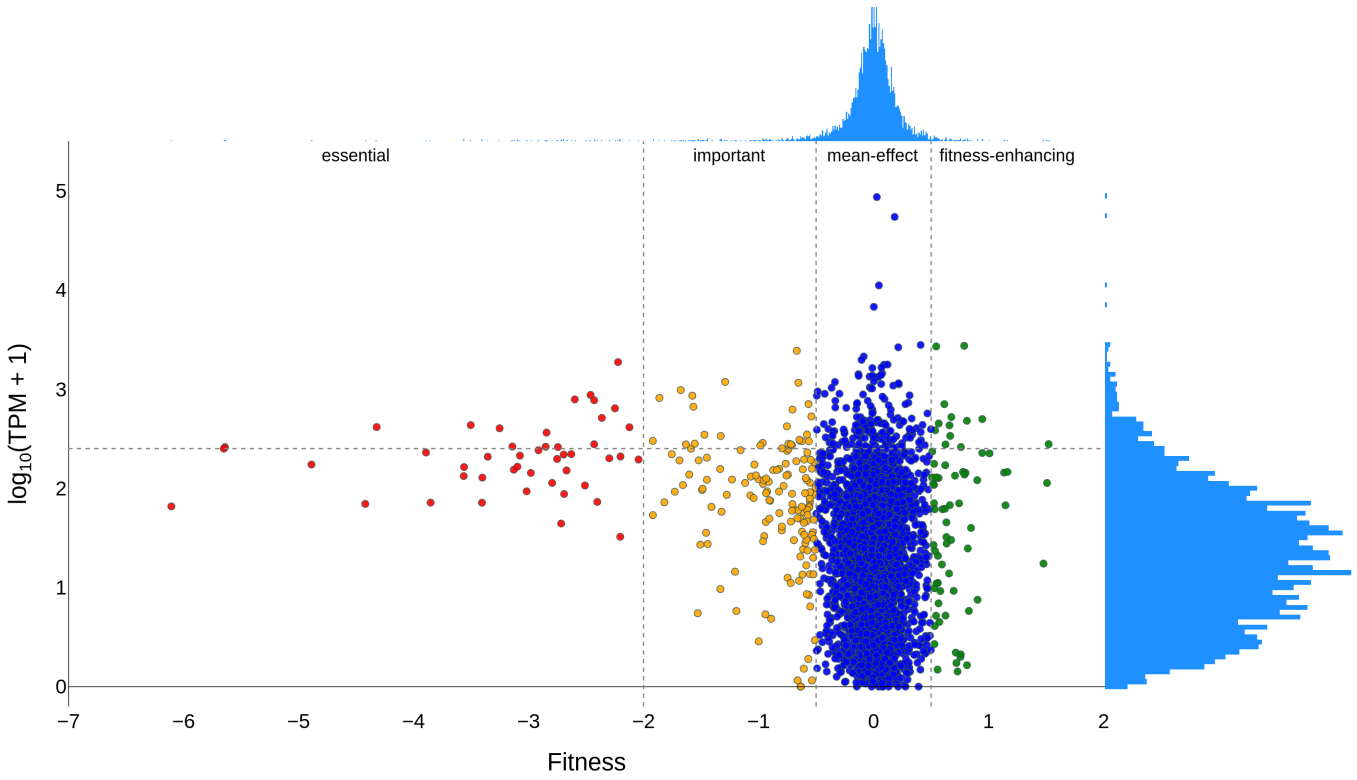
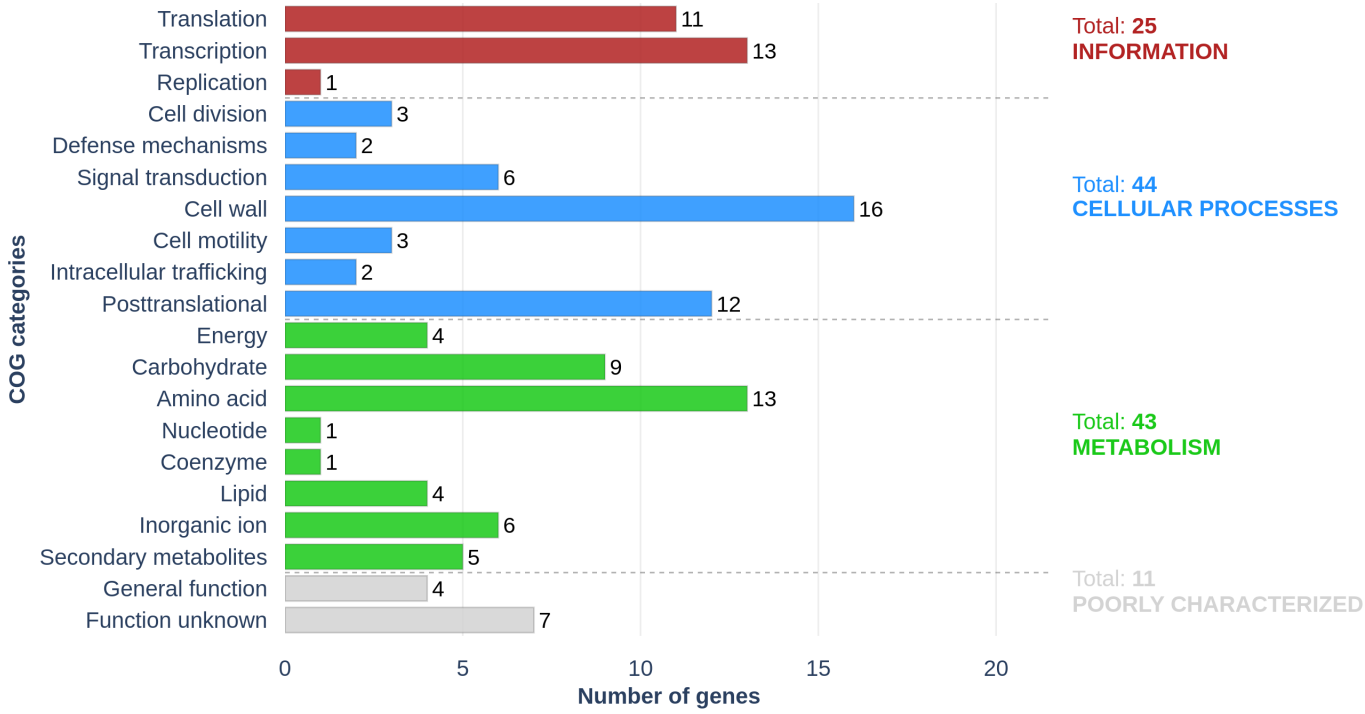


(A) Fitness under glucose (set6IT057) versus expression (log10(TPM+1), mean of three control replicates, GSE85475) for 4,477 genes. Colors indicate fitness categories: essential (≤ –2), important (–2 to –0.5), mean-effect (–0.5 to 0.5) and fitness-enhancing (> 0.5). Dashed horizontal line: top 5% expression cutoff (log10(TPM+1) ≥ 2.40).



(B) COG functional categories of the 168 highly expressed genes in the mean-effect category. Only genes with a single COG category are shown (123); 43 genes without COG assignment and 2 with multiple categories (K,H; M,N) are listed in the table (page 2).

Table. 168 highly expressed genes (top 5%, log10(TPM+1) ≥ 2.40) in the mean-effect category (fitness within ±0.5) under glucose

locusId	Symbol	Name	COG	Fitness	log10(TPM+1)
PP_0086		conserved protein of unknown function	R	0.227	2.637
PP_0168		putative surface adhesion protein	Q	0.307	2.533
PP_0181		conserved protein of unknown function		0.039	2.691
PP_0191	algQ	Transcriptional regulatory protein AlgQ	K	0.281	2.508
PP_0194	algP	Transcriptional regulatory protein AlgP		-0.234	2.414
PP_0234	oprE	outer-membrane porin E		-0.164	2.588
PP_0268	oprQ	outer-membrane porin D		0.032	2.680
PP_0282	artJ	L-arginine ABC transporter - periplasmic binding s…	T	0.197	2.524
PP_0319		conserved protein of unknown function	D	0.046	2.678
PP_0367		conserved protein of unknown function		-0.147	2.622
PP_0373	yqaE	conserved membrane protein of unknown function	C	-0.073	2.637
PP_0380	pqqA	Coenzyme PQQ synthesis protein A		-0.035	3.126
PP_0428	hinT	aminoacyl-adenylate hydrolase/purine nucleoside …	F	0.248	2.679
PP_0440	tufA	Elongation factor Tu-A	J	-0.086	3.328
PP_0482	bfr-I	bacterioferritin	P	0.211	2.709
PP_0621		conserved protein of unknown function		-0.180	2.503
PP_0641		conserved protein of unknown function		0.137	2.582
PP_0679		conserved exported protein of unknown function		0.022	3.117
PP_0684	fklB-I	FKBP-type peptidyl-prolyl cis-trans isomerase	O	0.016	2.431
PP_0697	rimI	ribosomal-protein-S18-alanine N-acetyltransferase	J	-0.311	2.435
PP_0750		conserved protein of unknown function		-0.336	3.071
PP_0753		putative Lipoprotein	S	0.337	2.545
PP_0834	yajC	Sec translocation complex subunit YajC	U	-0.013	2.573
PP_0848	iscX	donor of Fe2+ and regulator of cysteine desulfura…	O	0.468	2.755
PP_0850	rlmN	Dual-specificity RNA methyltransferase RlmN	J	-0.011	2.794
PP_0855		conserved protein of unknown function	T	-0.254	2.568
PP_0864	speC	ornithine decarboxylase	E	0.045	2.502
PP_0951	hpf	Ribosome hibernation promoting factor	J	-0.011	3.212
PP_0960	mlaD	phospholipid ABC transporter binding protein	M	0.412	2.486
PP_0961	ttg2D	toluene tolerance protein	M	0.373	2.606
PP_0963	ttg2F	putative BolA family protein	T	0.147	2.578
PP_1009	gapA	glyceraldehyde-3-phosphate dehydrogenase	G	-0.425	2.952
PP_1011	glk	glucokinase	G	-0.134	2.425
PP_1014	yihS	aldose-ketose isomerase	G	0.104	2.460
PP_1015	gtsA	mannose/glucose ABC transporter, glucose-bindi…	G	0.091	3.246
PP_1016	gtsB	mannose/glucose ABC transporter, permease pro…	G	-0.085	2.631
PP_1017	gtsC	mannose/glucose ABC transporter, permease pro…	G	0.030	2.663
PP_1018	gtsD	mannose/glucose ABC transporter - ATP binding …	G	0.147	2.742
PP_1019	oprB-I	carbohydrate-selective porin	M	-0.033	3.017
PP_1038		conserved protein of unknown function	S	-0.014	2.555
PP_1071	gltI	glutamate / aspartate ABC transporter - periplasm…	T	-0.146	2.849
PP_1082	bfr-II	bacterioferritin	P	0.043	3.125
PP_1084	tsaA	putative peroxiredoxin	V	-0.105	3.295
PP_1099	capB	Cold shock protein CapB	K	0.046	4.046
PP_1111		putative Synthetase	J	-0.271	2.471
PP_1114		SEC-C domain protein	S	0.319	2.574
PP_1131	slyB	outer membrane lipoprotein	M	-0.110	2.506
PP_1141	livK	branched-chain amino acids ABC transporter - pe…	E	-0.080	2.685
PP_1185	oprH	Outer membrane protein H1		0.302	2.863
PP_1206	oprD	basic-amino-acid specific porin OprD		-0.297	2.953
PP_1208		conserved protein of unknown function, SlyX family	S	-0.357	2.527
PP_1218		putative acyl-CoA thioesterase	I	0.335	2.586
PP_1224	ybgF	periplasmic subunit of the TolQRA transport system	D	-0.025	3.008
PP_1238		putative Lipoprotein	M	-0.079	2.408
PP_1239		Metallo-beta-lactamase family protein	P	0.066	2.552
PP_1245		conserved exported protein of unknown function	M	0.168	2.417
PP_1249		conserved protein of unknown function, DUF4223…		0.183	4.736
PP_1303	cysD	Sulfate adenylyltransferase subunit 2	E	-0.433	2.647
PP_1304	cysNC	sulfate adenylyltransferase subunit 1/adenylyl-sulf…	P	-0.493	2.590
PP_1319	petC	Ubiquinol--cytochrome c reductase, cytochrome c1	C	-0.365	3.013
PP_1328	mraZ	Protein MraZ	J	0.276	2.848
PP_1352		UPF0234 protein PP_1352	T	-0.006	2.532
PP_1366		Transcriptional regulator MvaT, P16 subunit		0.409	3.445
PP_1427	rpoE	RNA polymerase sigma E factor	K	0.032	2.688
PP_1522	cspA-I	major cold shock protein	K	0.002	3.830
PP_1576		putative Immunity protein		-0.007	2.766
PP_1609		conserved protein of unknown function		-0.132	3.150
PP_1611	kdsA-I	2-dehydro-3-deoxyphosphooctonate aldolase	M	-0.125	2.451
PP_1623	rpoS	RNA polymerase, sigma S (sigma 38) factor	K	0.426	2.639
PP_1663		conserved exported protein of unknown function		0.222	2.424
PP_1714	fklB-II	FKBP-type peptidyl-prolyl cis-trans isomerase	O	-0.341	2.576
PP_1716		Transcriptional regulator, Cro/CI family	K	-0.192	2.446
PP_1733	minD	membrane ATPase of the MinC-MinD-MinE system	D	0.319	2.508
PP_1757	bolA	DNA-binding transcriptional dual regulator	K	0.099	2.419
PP_1782	rfbC	dTDP-4-dehydrorhamnose 3,5-epimerase	M	-0.077	2.506
PP_1783	rfbA	dTDP-glucose pyrophosphorylase	M	-0.108	2.610
PP_1786		putative Glycosyl transferase	M,N	0.164	2.410
PP_1788		conserved protein of unknown function		-0.333	2.882
PP_1790		putative Acylneuraminate cytidylyltransferase	M	-0.394	2.598
PP_1791		putative Aldolase/synthase	E	-0.006	2.462
PP_1794		conserved hypothetical protein	Q	-0.010	3.135
PP_1795		conserved protein of unknown function		-0.012	2.754
PP_1799	gmd	GDP-mannose 4,6-dehydratase	M	0.410	2.566
PP_1982	ibpA	small heat shock protein IbpA		-0.097	2.682
PP_2082	ppsA	phosphoenolpyruvate synthetase	G	0.109	2.449
PP_2088		RNA polymerase sigma factor SigX	K	-0.017	2.589
PP_2099		conserved protein of unknown function	S	-0.151	2.800
PP_2118		conserved protein of unknown function		0.313	2.937
PP_2121		putative Lipoprotein		-0.143	2.860
PP_2220		C4-type zinc finger protein, DksA/TraR family	K	0.247	2.455
PP_2294		conserved exported protein of unknown function		-0.092	2.946
PP_2299	tig	Trigger factor	O	-0.494	2.933
PP_2302	lon-II	DNA-binding, ATP-dependent protease	O	-0.384	2.461
PP_2322	oprI	Major outer membrane lipoprotein		0.028	4.936
PP_2324	aroF-I	tyr-sensitive phospho-2-dehydro-3-deoxyheptonat…	E	-0.025	2.565
PP_2472		putative transcriptional regulator, MerR family	K	-0.333	2.815
PP_2550		putative transcriptional regulatory protein	J	-0.352	2.574
PP_2947		putative Transcriptional regulator MvaT, P16 subunit		0.138	2.474
PP_2969	csoR	putative repressor of copper utilisation proteins	K	0.247	2.424
PP_3024		conserved protein of unknown function		0.071	2.496
PP_3035		conserved protein of unknown function		0.214	3.421
PP_3089		conserved protein of unknown function	U	0.074	3.160
PP_3376	ptxD	putative phosphonate dehydrogenase	C	-0.100	2.578
PP_3407		conserved protein of unknown function		0.498	2.595
PP_3417	gntT	D-gluconate transporter	G	0.002	3.078
PP_3504		conserved protein of unknown function		0.312	2.598
PP_3587	tpx	lipid hydroperoxide peroxidase	O	0.140	2.492
PP_3682		conserved protein of unknown function		-0.045	2.403
PP_3693		putative Transcriptional regulator MvaT, P16 subunit		-0.145	2.530
PP_3775		putative sarcosine oxidase	E	-0.021	2.640
PP_3776		putative permease (RarD protein)	R	-0.059	2.607
PP_3777		conserved protein of unknown function	J	0.150	2.791
PP_3778	proC	Pyrroline-5-carboxylate reductase	E	0.095	2.900
PP_3781		Oxygen-independent Coproporphyrinogen III oxid…	H	0.065	3.141
PP_3782		conserved protein of unknown function		0.121	3.247
PP_3783	syrB	syringomycin biosynthesis enzyme 2		0.068	3.218
PP_3784		conserved protein of unknown function	V	0.170	3.035
PP_3785		conserved protein of unknown function		-0.051	2.881
PP_3786	aspC	aminotransferase	E	-0.121	2.656
PP_3787		conserved protein of unknown function	I	-0.070	2.636
PP_3788		putative Non-ribosomal peptide synthetase	Q	0.026	2.509
PP_3821	galU	UTP-glucose-1-phosphate uridylyltransferase	M	-0.114	2.718
PP_3901		conserved protein of unknown function		0.183	2.402
PP_3904		conserved protein of unknown function		0.220	3.049
PP_3909		conserved protein of unknown function		-0.026	2.830
PP_3924		conserved protein of unknown function	I	-0.146	2.718
PP_3963		conserved protein of unknown function	R	-0.242	2.456
PP_3990	yccA	substrate or modulator of FtsH-mediated proteolysis	P	-0.030	2.690
PP_4008	clpA	ATP-dependent serine protease	O	0.014	2.528
PP_4009	clpS	ATP-dependent Clp protease adapter protein ClpS	O	-0.253	2.553
PP_4179	htpG	Chaperone protein HtpG	O	-0.155	2.638
PP_4268	ybaB	nucleoid-associated broad specificity regulator	K	-0.130	3.133
PP_4375	fliS	flagellar chaperone, inhibits premature FliC asse…	N	-0.035	2.405
PP_4376	fliD	flagellar filament capping protein	N	-0.028	2.559
PP_4378	fliC	flagellin, filament structural protein	N	-0.013	3.006
PP_4502	yciI	putative enzyme	Q	0.445	2.435
PP_4519	tolC	Agglutination protein	M	0.331	2.703
PP_4569		conserved protein of unknown function		-0.359	2.536
PP_4593		conserved protein of unknown function		-0.353	2.453
PP_4640		conserved protein of unknown function	S	0.079	2.564
PP_4645	mscL	Large-conductance mechanosensitive channel	M	0.113	2.528
PP_4720		putative RNA binding protein	J	-0.486	2.976
PP_4721		conserved membrane protein of unknown function		0.075	2.653
PP_4722	greA	Transcription elongation factor GreA	K	0.131	2.836
PP_4775		conserved protein of unknown function		0.066	3.051
PP_4803	dacA	D-alanyl-D-alanine carboxypeptidase	M	0.175	2.690
PP_4820	dusB	tRNA-dihydrouridine synthase B	J	0.000	2.509
PP_4835		conserved exported protein of unknown function		-0.261	2.593
PP_4891	hflC	integral membrane ATP-dependent zinc metallop…	O	-0.360	2.667
PP_4892	hflK	integral membrane ATP-dependent zinc metallop…	O	-0.049	2.577
PP_4893	hflX	GTPase HflX	J	-0.121	2.567
PP_4911		conserved protein of unknown function	M	0.059	2.655
PP_4966	sahR	transcriptional regulator of methionine metabolism	K,H	0.094	2.655
PP_4977	metF	5,10-methylenetetrahydrofolate reductase	E	-0.070	2.428
PP_4981	yceI	polyisoprene-binding protein, acidic stress respon…	R	-0.093	2.675
PP_5024		Amino acid ABC transporter, periplasmic amino a…	T	0.106	2.634
PP_5039		conserved protein of unknown function	S	-0.022	2.650
PP_5054	grxC	glutaredoxin 3	O	0.394	2.464
PP_5055		Sulfurtransferase	P	-0.126	2.580
PP_5075	gltD	glutamate synthase (NADPH) beta subunit	E	-0.417	2.513
PP_5076	gltB	L-glutamate synthase(NADPH) alpha subunit	E	-0.132	2.574
PP_5085	maeB	malic enzyme B	C	0.216	3.058
PP_5181	spuD	spermidine/putrescine ABC transporter - periplas…	E	-0.008	2.573
PP_5193	gcvH-II	glycine cleavage system H protein 2	E	-0.249	2.482
PP_5204		conserved protein of unknown function		-0.238	2.812
PP_5313	hupA	DNA-binding protein HU-alpha	L	0.260	2.673
PP_5353		conserved exported protein of unknown function	I	0.173	2.494
PP_5395		conserved protein of unknown function	Q	0.077	2.924